Exponential and Logarithmic Equations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponential equations**

Solve $3^{x+1} = 81$ exactly.

$3^{x+1} = 3^4$, so $x + 1 = 4$ and $x = 3$.

2 **[Foundational]** • **Target: Exponential equations**

Solve $5^{2x} = 17$. Give an exact answer and a decimal to three places.

$2x = \log_5 17 = \frac{\ln 17}{\ln 5}$, so $x = \frac{\ln 17}{2\ln 5} \approx 0.880$.

3 **[Foundational]** • **Target: Exponential equations**

Solve $e^{2x} - 5e^x + 6 = 0$ exactly.

Let $u = e^x$: $u^2 - 5u + 6 = (u - 2)(u - 3) = 0$, so $e^x = 2$ or $e^x = 3$: $x = \ln 2$ or $x = \ln 3$.

4 **[Foundational]** • **Target: Logarithmic equations**

Solve $\ln(2x - 3) = 1$. Give an exact answer and a decimal to three places.

$2x - 3 = e$, so $x = \frac{e+3}{2} \approx 2.859$.

5 **[Foundational]** • **Target: Logarithmic equations**

Solve $\log_2 x + \log_2 (x - 6) = 4$, checking for extraneous solutions.

$\log_2 (x(x - 6)) = 4$, so $x^2 - 6x - 16 = 0$, giving $(x - 8)(x + 2) = 0$. Reject $x = -2$ (logarithm of a negative number): $x = 8$.

6 **[Foundational]** • **Target: Logarithmic equations**

Solve $\log(x + 3) - \log(x - 1) = 1$, checking for extraneous solutions.

$\frac{x+3}{x-1} = 10$, so $x + 3 = 10x - 10$ and $x = \frac{13}{9}$. Check: $x > 1$, valid.

7 **[Foundational]** • **Target: Logarithmic equations**

The pH of a solution is $\text{pH} = -\log[\text{H}^+]$. Find the hydrogen-ion concentration $[\text{H}^+]$ of a solution with pH 4.6, in scientific notation to two significant figures.

$[\text{H}^+] = 10^{-4.6} \approx 2.5 \times 10^{-5}$ mol/L.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More exponential equations

Solve $2^x = 3^{x-1}$, giving an exact expression and a decimal to three places.

Take logs: $x\ln 2 = (x-1)\ln 3$, so $x(\ln 2 - \ln 3) = -\ln 3$, giving $x = \frac{\ln 3}{\ln 3 - \ln 2} = \frac{\ln 3}{\ln(3/2)} \approx \frac{1.0986}{0.4055} \approx 2.710$.

9 [Intermediate] • Target: More exponential equations

Solve $4^x - 3 \cdot 2^x - 4 = 0$ exactly, using the substitution $u = 2^x$.

$4^x = (2^x)^2 = u^2$: $u^2 - 3u - 4 = 0$, so $(u-4)(u+1) = 0$. $u = 4$ (rejecting $u = -1$, since $2^x > 0$): $2^x = 4$, so $x = 2$.

10 [Intermediate] • Target: Systems and applications

A bacteria culture grows according to $N(t) = 500e^{kt}$. If the population reaches 2000 after 6 hours, find k (4 d.p.) and hence the time for the population to reach 10,000, to one decimal place.

$2000 = 500e^{6k}$: $e^{6k} = 4$, so $k = \frac{\ln 4}{6} \approx 0.2310$. For $N = 10000$: $10000 = 500e^{0.2310t}$, so $e^{0.2310t} = 20$, $t = \frac{\ln 20}{0.2310} \approx 12.97 \approx 13.0$ hours.

11 [Intermediate] • Target: Systems and applications

Solve the equation $e^x + e^{-x} = 3$ exactly, using the substitution $u = e^x$ (this expression is related to $2\cosh x$).

Multiply by $u = e^x$: $u^2 - 3u + 1 = 0$, so $u = \frac{3 \pm \sqrt{5}}{2}$. Both roots are positive, giving $x = \ln\left(\frac{3+\sqrt{5}}{2}\right)$ or $x = \ln\left(\frac{3-\sqrt{5}}{2}\right)$.

12 [Intermediate] • Target: Combined logarithmic and exponential practice

Solve $\log_3(x+2) + \log_3(x-2) = 2$, checking for extraneous solutions.

$\log_3((x+2)(x-2)) = 2$: $x^2 - 4 = 9$, so $x^2 = 13$, $x = \pm\sqrt{13}$. Reject $x = -\sqrt{13}$ (makes $x - 2 < 0$): $x = \sqrt{13}$.

13 [Intermediate] • Target: Combined logarithmic and exponential practice

Solve $9^x = 27^{x-1}$ exactly, writing both sides with base 3.

$3^{2x} = 3^{3(x-1)}$: $2x = 3x - 3$, so $x = 3$.

14 [Intermediate] • Target: Equations requiring both exponential and logarithmic techniques

Solve $\log_2(x) = 3 - \log_2(x-2)$, checking for extraneous solutions.

$\log_2 x + \log_2(x-2) = 3 \Rightarrow \log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Reject $x = -2$ (negative argument): $x = 4$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Equations requiring both exponential and logarithmic techniques** • **Cross-Lesson: 3.01 The Unit Circle and Exact Values**

Solve $e^{\ln x + \ln 2} = 10$ for x, simplifying the exponent first.

$e^{\ln x + \ln 2} = e^{\ln(2x)} = 2x$ (since $e^{\ln u} = u$). So $2x = 10 \Rightarrow x = 5$.

16 **[Challenging]** • **Target: Verifying solutions numerically** • **Cross-Lesson: 3.02 Fundamental Trigonometric Identities**

For the equation $5^{2x} = 17$ solved earlier ($x \approx 0.880$), verify the solution by substituting back and computing both sides to 2 decimal places.

$5^{2(0.880)} = 5^{1.76} \approx 17.02$, matching the right side 17 (small rounding difference due to the decimal approximation of x), confirming the solution.

17 **[Challenging]** • **Target: Equations with different bases on each side** • **Cross-Lesson: 3.03 Sum and Difference Formulas**

Solve $7^x = 3^{x+2}$, giving an exact expression and a decimal to three places.

Take logs: $x\ln 7 = (x+2)\ln 3$: $x(\ln 7 - \ln 3) = 2\ln 3$, so $x = \frac{2\ln 3}{\ln 7 - \ln 3} = \frac{2\ln 3}{\ln(7/3)} \approx \frac{2.1972}{0.8473} \approx 2.593$.

18 **[Challenging]** • **Target: Equations with different bases on each side** • **Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations**

Solve $10^{2x-1} = 500$, giving an exact logarithmic expression and a decimal to three places.

$2x - 1 = \log 500$, so $x = \frac{1 + \log 500}{2} \approx \frac{1 + 2.6990}{2} \approx 1.850$.

19 **[Challenging]** • **Target: A modeling problem requiring both equation types** • **Cross-Lesson: 4.01 Circles and Parabolas**

A population model is $P(t) = 800e^{0.05t}$. Find the time t when $P(t) = 2000$, to two decimal places, and separately find $P^{-1}(1500)$ (the time when the population reaches 1500), to two decimal places.

For $P = 2000$: $2000 = 800e^{0.05t} \Rightarrow 2.5 = e^{0.05t} \Rightarrow t = \frac{\ln 2.5}{0.05} \approx 18.33$. For $P = 1500$: $1500 = 800e^{0.05t} \Rightarrow 1.875 = e^{0.05t} \Rightarrow t = \frac{\ln 1.875}{0.05} \approx 12.58$.

20 **[Challenging]** • **Target: Checking solutions by substitution** • **Cross-Lesson: 4.02 Ellipses**

Verify the solution $x = 8$ found earlier for $\log_2 x + \log_2 (x - 6) = 4$ by substituting back into the original equation.

$\log_2 (8) + \log_2 (8 - 6) = \log_2 (8) + \log_2 (2) = 3 + 1 = 4$ ✓, confirming $x = 8$ satisfies the original equation.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Equations with Like Bases

Solve $27^{x-1} = 9^{2x+3}$ for x.

a. -9 ✓

b. 9

c. -3

d. 6

Base 3: $(3^3)^{x-1} = (3^2)^{2x+3} \Rightarrow 3(x-1) = 2(2x+3) \Rightarrow 3x-3 = 4x+6 \Rightarrow -x = 9 \Rightarrow x = -9$.

22 [Foundational] • Target: Equations with Like Bases

Solve $5^{x^2-3x} = 625$.

a. $x = -1$ **and** $x = 4$ ✓

b. $x = 1$ and $x = -4$

c. $x = 2$ and $x = -2$

d. $x = 4$ only

$625 = 5^4 \Rightarrow x^2 - 3x = 4 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x-4)(x+1) = 0 \Rightarrow x = 4, -1$.

23 [Foundational] • Target: Equations with Like Bases

Solve $\left(\frac{1}{2}\right)^{x+2} = 8^{x-1}$.

a. $1/4$ ✓

b. $1/2$

c. $-1/4$

d. 2

$2^{-(x+2)} = 2^{3(x-1)} \Rightarrow -x-2 = 3x-3 \Rightarrow 4x = 1 \Rightarrow x = 1/4$.

24 [Foundational] • Target: Equations Requiring Natural Logarithms

Solve $3^{2x} = 7$ for x in exact form.

a. $x = \frac{\ln 7}{2\ln 3}$ ✓

b. $x = \frac{\ln 3}{2\ln 7}$

c. $x = \frac{7}{6}$

d. $x = \frac{\ln 7}{\ln 6}$

Take $\ln$ of both sides: $\ln(3^{2x}) = \ln 7 \Rightarrow 2x\ln 3 = \ln 7 \Rightarrow x = \frac{\ln 7}{2\ln 3}$.

25 [Foundational] • Target: Equations Requiring Natural Logarithms

Solve $5e^{3x-2} = 40$ for x.

a. $x = \frac{\ln 8 + 2}{3}$ ✓

b. $x = \frac{\ln 8 - 2}{3}$

c. $x = \frac{\ln 40}{15}$

d. $x = 2$

$e^{3x-2} = 8 \Rightarrow 3x - 2 = \ln 8 \Rightarrow 3x = \ln 8 + 2 \Rightarrow x = \frac{\ln 8 + 2}{3}$.

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26 [Foundational] • Target: Equations Requiring Natural Logarithms

Solve $2^{x+1} = 5^{x-1}$ for x.

a. $x = \frac{\ln 2 + \ln 5}{\ln 5 - \ln 2} = \frac{\ln 10}{\ln(5/2)}$ ✓

b. $x = \frac{\ln 5}{\ln 2}$

c. $x = 1$

d. $x = \frac{\ln 10}{\ln 10} = 1$

$(x+1)\ln 2 = (x-1)\ln 5 \Rightarrow x\ln 2 + \ln 2 = x\ln 5 - \ln 5 \Rightarrow x(\ln 5 - \ln 2) = \ln 5 + \ln 2 \Rightarrow x = \frac{\ln 10}{\ln(5/2)}$.

27 [Foundational] • Target: Logarithmic Equations and Extraneous Solutions

Solve $\log_2(x) + \log_2(x-2) = 3$.

a. $x = 4$ **only** ✓

b. $x = 4$ and $x = -2$

c. $x = 8$

d. $x = 2$

$\log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 2^3 = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Since $\log_2(-2)$ is undefined, $x = -2$ is extraneous. Solution: $x = 4$.

28 [Foundational] • Target: Logarithmic Equations and Extraneous Solutions

Solve $\ln(x+5) - \ln(x-1) = \ln(3)$.

a. $x = 4$ ✓

b. $x = 3$

c. $x = 2$

d. $x = 5$

$\ln\left(\frac{x+5}{x-1}\right) = \ln(3) \Rightarrow \frac{x+5}{x-1} = 3 \Rightarrow x + 5 = 3x - 3 \Rightarrow 2x = 8 \Rightarrow x = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Logarithmic Equations and Extraneous Solutions

Why must potential solutions to logarithmic equations always be checked in the original equation?

a. **The argument of every logarithm must be strictly positive (> 0)** ✓

b. Logarithms cannot have rational answers

c. Extraneous roots always equal 0

d. Base change invalidates solutions

Logarithmic functions are only defined for positive real inputs; algebraic manipulation can introduce extraneous roots that make arguments ≤ 0.

30 [Intermediate] • Target: Quadratic-Form Exponential Equations

Solve $e^{2x} - 5e^x + 6 = 0$ for x.

a. $x = \ln 2$ **and** $x = \ln 3$ ✓

b. $x = 2$ and $x = 3$

c. $x = \ln 5$ and $x = \ln 6$

d. $x = e^2$ and $x = e^3$

Let $u = e^x$: $u^2 - 5u + 6 = (u - 2)(u - 3) = 0 \Rightarrow u = 2, 3$. Thus $x = \ln 2$ or $x = \ln 3$.

31 [Intermediate] • Target: Quadratic-Form Exponential Equations

Solve $e^{2x} - 3e^x - 4 = 0$ for x.

a. $x = \ln 4$ **only** ✓

b. $x = \ln 4$ and $x = \ln(-1)$

c. $x = 4$ and $x = -1$

d. No real solutions

$(e^x - 4)(e^x + 1) = 0 \Rightarrow e^x = 4$ or $e^x = -1$. Since $e^x > 0$ for all real x, $e^x = -1$ has no real solution. Thus $x = \ln 4$.

32 [Intermediate] • Target: Quadratic-Form Exponential Equations

Solve $4^x - 3 \cdot 2^x - 4 = 0$.

a. $x = 2$ **only** ✓

b. $x = 4$ only

c. $x = 2$ and $x = -1$

d. $x = 1$

$(2^x)^2 - 3(2^x) - 4 = 0 \Rightarrow (2^x - 4)(2^x + 1) = 0 \Rightarrow 2^x = 4 \Rightarrow x = 2$ (since $2^x = -1$ is impossible).

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33 **[Intermediate]** • **Target: Logarithmic Equations and Extraneous Solutions**

Solve $7^x = 100$ for x, expressed using logarithms.

a. $x = \frac{100}{7}$

b. $x = 100 - 7$

c. $x = \log_7(100) = \frac{\ln 100}{\ln 7}$ ✓

d. $x = \log_{100}(7)$

Take $\ln$ of both sides: $x \ln 7 = \ln 100$, so $x = \frac{\ln 100}{\ln 7} \approx 2.367$.

34 **[Intermediate]** • **Target: Quadratic-Form Exponential Equations**

Solve $7^x = 10$ for x, expressed using logarithms.

a. $x = 10 - 7$

b. $x = \log_7(10) = \frac{\ln 10}{\ln 7}$ ✓

c. $x = \frac{10}{7}$

d. $x = \log_{10}(7)$

Take $\ln$ of both sides: $x \ln 7 = \ln 10$, so $x = \frac{\ln 10}{\ln 7} \approx 1.183$.

35 **[Intermediate]** • **Target: Exponential equations**

Solve $3^x = 50$ for x, expressed using logarithms.

a. $x = \log_{50}(3)$

b. $x = \frac{50}{3}$

c. $x = \log_3(50) = \frac{\ln 50}{\ln 3}$ ✓

d. $x = 50 - 3$

Take $\ln$ of both sides: $x \ln 3 = \ln 50$, so $x = \frac{\ln 50}{\ln 3} \approx 3.561$.

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36 [Intermediate] • Target: Logarithmic equations

Solve $5^x = 20$ for x, expressed using logarithms.

a. $x = \log_5(20) = \frac{\ln 20}{\ln 5}$ ✓

b. $x = \frac{20}{5}$

c. $x = \log_{20}(5)$

d. $x = 20 - 5$

Take $\ln$ of both sides: $x \ln 5 = \ln 20$, so $x = \frac{\ln 20}{\ln 5} \approx 1.861$.

37 [Intermediate] • Target: More exponential equations

Solve $2^x = 15$ for x, expressed using logarithms.

a. $x = \frac{15}{2}$

b. $x = 15 - 2$

c. $x = \log_2(15) = \frac{\ln 15}{\ln 2}$ ✓

d. $x = \log_{15}(2)$

Take $\ln$ of both sides: $x \ln 2 = \ln 15$, so $x = \frac{\ln 15}{\ln 2} \approx 3.907$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Systems and applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Solve $3^x = 20$ for x, expressed using logarithms.

a. $x = \log_{20}(3)$

b. $x = \frac{20}{3}$

c. $x = 20 - 3$

d. $x = \log_3(20) = \frac{\ln 20}{\ln 3}$ ✓

Take $\ln$ of both sides: $x \ln 3 = \ln 20$, so $x = \frac{\ln 20}{\ln 3} \approx 2.727$.

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39 [Challenging] • Target: Combined logarithmic and exponential practice • Cross-Lesson: 4.01 Circles and Parabolas

Solve $2^x = 20$ for x, expressed using logarithms.

a. $x = \log_{20}(2)$

b. $x = 20 - 2$

c. $x = \frac{20}{2}$

d. $x = \log_2(20) = \frac{\ln 20}{\ln 2}$ ✓

Take $\ln$ of both sides: $x \ln 2 = \ln 20$, so $x = \frac{\ln 20}{\ln 2} \approx 4.322$.

40 [Challenging] • Target: Equations requiring both exponential and logarithmic techniques • Cross-Lesson: 4.02 Ellipses

Solve $7^x = 15$ for x, expressed using logarithms.

a. $x = \frac{15}{7}$

b. $x = 15 - 7$

c. $x = \log_{15}(7)$

d. $x = \log_7(15) = \frac{\ln 15}{\ln 7}$ ✓

Take $\ln$ of both sides: $x \ln 7 = \ln 15$, so $x = \frac{\ln 15}{\ln 7} \approx 1.392$.

41 [Challenging] • Target: Verifying solutions numerically • Cross-Lesson: 4.03 Hyperbolas

Solve $3^x = 10$ for x, expressed using logarithms.

a. $x = \log_3(10) = \frac{\ln 10}{\ln 3}$ ✓

b. $x = 10 - 3$

c. $x = \log_{10}(3)$

d. $x = \frac{10}{3}$

Take $\ln$ of both sides: $x \ln 3 = \ln 10$, so $x = \frac{\ln 10}{\ln 3} \approx 2.096$.

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42 [Challenging] • Target: Equations with different bases on each side • Cross-Lesson: 5.01 Vectors: Components and Operations

Solve $3^x = 15$ for x, expressed using logarithms.

a. $x = \log_{15}(3)$

b. $x = \frac{15}{3}$

c. $x = \log_3(15) = \frac{\ln 15}{\ln 3}$ ✓

d. $x = 15 - 3$

Take $\ln$ of both sides: $x\ln 3 = \ln 15$, so $x = \frac{\ln 15}{\ln 3} \approx 2.465$.

43 [Challenging] • Target: A modeling problem requiring both equation types • Cross-Lesson: 5.02 The Dot Product

Solve $7^x = 50$ for x, expressed using logarithms.

a. $x = \frac{50}{7}$

b. $x = \log_7(50) = \frac{\ln 50}{\ln 7}$ ✓

c. $x = \log_{50}(7)$

d. $x = 50 - 7$

Take $\ln$ of both sides: $x\ln 7 = \ln 50$, so $x = \frac{\ln 50}{\ln 7} \approx 2.01$.

44 [Challenging] • Target: Checking solutions by substitution • Cross-Lesson: 5.03 Parametric Equations

Write $7^4 = 2401$ in logarithmic form.

a. $\log_7(4) = 2401$

b. $\log_7(2401) = 4$ ✓

c. $\log_4(7) = 2401$

d. $\log_{2401}(7) = 4$

$b^x = y \Leftrightarrow \log_b(y) = x$, so $\log_7(2401) = 4$.

45 [Challenging] • Target: Equations with Like Bases • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Write $5^4 = 625$ in logarithmic form.

a. $\log_{625}(5) = 4$

b. $\log_5(4) = 625$

c. $\log_4(5) = 625$

d. $\log_5(625) = 4$ ✓

$b^x = y \Leftrightarrow \log_b(y) = x$, so $\log_5(625) = 4$.

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